

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[2\sqrt{3} \left(\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[\sqrt{3} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2. $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $\left[\sqrt{2} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4. $x^4 + 4 = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = -i$	YES / NO	b) $z = 2i$	YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!*Every line shown. Point splits are an estimate of partial credit (halves are possible).*

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[2\sqrt{3} \left(\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[\sqrt{3} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$

1. Multiply the r 's, add the θ 's (common denominator first).

$$r = (2\sqrt{3})(\sqrt{3}) = 2 \cdot 3 = 6 \quad \theta = \frac{13\pi}{12} + \frac{15\pi}{12} = \frac{28\pi}{12} = \frac{7\pi}{3} \xrightarrow{-2\pi} \frac{\pi}{3}$$

$$6 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 6 \left(\frac{1}{2} \right) + 6i \left(\frac{\sqrt{3}}{2} \right) = \boxed{3 + 3\sqrt{3}i}$$

Trap: $\frac{13\pi}{12}$ is not on the unit circle, but the sum is. Add first, then look.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

$$\text{Top: } r = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4 \quad \tan \theta = \frac{-2\sqrt{3}}{-2} = \sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIII} \Rightarrow 240^\circ$$

$$\text{Bottom: } r = \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{8} = 2\sqrt{2} \quad \tan \theta = \frac{-\sqrt{6}}{\sqrt{2}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$$

$$\frac{4 \text{cis } 240^\circ}{2\sqrt{2} \text{cis } 300^\circ} = \frac{2}{\sqrt{2}} \text{cis}(-60^\circ) \xrightarrow{+360^\circ} \sqrt{2} \text{cis } 300^\circ = \sqrt{2} \left(\frac{1}{2} \right) + \sqrt{2}i \left(-\frac{\sqrt{3}}{2} \right) = \boxed{\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}i}$$

Trap: $\frac{2}{\sqrt{2}} = \sqrt{2}$. Both parts of the top negative: QIII, 240° (not 60°).

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

3. De Moivre: raise r to the 4th, multiply the angle by 4, then reduce.

$$(\sqrt{2})^4 = 4 \quad 4 \cdot \frac{7\pi}{12} = \frac{28\pi}{12} = \frac{7\pi}{3} \xrightarrow{-2\pi} \frac{\pi}{3}$$

$$4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 4 \left(\frac{1}{2} \right) + 4i \left(\frac{\sqrt{3}}{2} \right) = \boxed{2 + 2\sqrt{3}i}$$

Trap: $\frac{7\pi}{12}$ is not a unit-circle angle, but $4 \cdot \frac{7\pi}{12}$ reduces to one. Multiply first, then reduce.

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

Problem: $x^4 + 4 = 0$

4. Isolate x^4 , write the constant in trig form, take fourth roots, convert each.

$$x^4 = -4 = -4 + 0i \Rightarrow r = 4, \theta = 180^\circ \quad x = (4 \operatorname{cis} 180^\circ)^{1/4} = \sqrt[4]{4} \operatorname{cis} \frac{180^\circ}{4} = \sqrt{2} \operatorname{cis} 45^\circ \quad + 90^\circ$$

$$\sqrt{2} \operatorname{cis} 45^\circ = \sqrt{2} \left(\frac{\sqrt{2}}{2} \right) + \sqrt{2} i \left(\frac{\sqrt{2}}{2} \right) = 1 + i \quad \sqrt{2} \operatorname{cis} 135^\circ = -1 + i$$

$$\sqrt{2} \operatorname{cis} 225^\circ = -1 - i \quad \sqrt{2} \operatorname{cis} 315^\circ = 1 - i$$

$$x = 1 + i, -1 + i, -1 - i, 1 - i$$

Trap: $\sqrt[4]{4} = \sqrt{2}$. $x = \pm\sqrt{2}i$ is wrong: $(\sqrt{2}i)^4 = 4$, not -4 . The roots sit between the axes.

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

Problem: a) $z = -i$

YES / NO

b) $z = 2i$

YES / NO

5. Keep computing until an absolute value goes over 2 (then NO) or the values start repeating (then YES).

a) $z = -i$: $|-i| = 1$

$$z^2 + z = (-i)^2 - i = -1 - i, \quad |-1 - i| = \sqrt{2} \approx 1.4$$

$$(z^2 + z)^2 + z = (-1 - i)^2 - i = (1 + 2i + i^2) - i = 2i - i = i, \quad |i| = 1$$

$$i^2 - i = -1 - i \text{ again: it repeats } (-1 - i, i, -1 - i, \dots) \text{ and never passes 2}$$

YES

b) $z = 2i$: $|2i| = 2$ (not over 2 yet, keep going)

$$z^2 + z = (2i)^2 + 2i = -4 + 2i, \quad |-4 + 2i| = \sqrt{16 + 4} = \sqrt{20} \approx 4.5 > 2$$

NO

Trap: $(-i)^2 = i^2 = -1$ and $(2i)^2 = 4i^2 = -4$. Exactly 2 is still in; only more than 2 is out.